# Meeting Brief

The blue revision of *Two Portable Components for Meeting Hard Prediction Quotas*

Prepared 9 August 2026, updated 13 August · read this one document; everything else is backup

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## 1. What you are presenting, in four sentences

The manuscript is unchanged in substance: same claims, same headline numbers, same structure. What is new is a body of follow-up work (Track B, roughly 600 runs) that answers the open questions the paper itself flagged, and five insertions that fold those answers in. Every insertion is blue. Every sentence the new work made obsolete is struck in red with a blue replacement beside it. Your professor can read the paper start to finish and accept the revision without editing a word of his own text.

**The single most important fact to state up front**

`paper/main.tex` — the version he wrote — is byte-identical to what he sent. The revision lives in a separate file. This was verified mechanically, not by eye: a checker strips every blue and red character and compares the remainder to his original at character level. It reports that not one character of his prose was added, removed, or altered. The only structural change is one figure relocated from §1 to Appendix E.

## 2. The five blue insertions

1. **§4, one clause** — notes that the native-resolution question is no longer entirely deferred. Sets up insertion 4.
2. **§5.1, one paragraph + Table 2** — three robustness checks: a stronger dual baseline (ALM), a fourth backbone (MobileNetV2), and interval estimates (bootstrap CIs, BH-FDR). *From B3, B8, B6, B7.*
3. **§5.3, one paragraph + Table 3** — the imbalanced-learning baselines (focal, class-balanced, logit adjustment), each trained with its own loss and then clipped. *From B1.*
4. **§7, one paragraph + Table 4** — a native-resolution check on HAM10000, one dataset run to full depth: every method, every backbone, two caps, four seeds, 96 runs. *From B2, reworked.*
5. **Ablation appendix, one paragraph** — the component ablation repeated in the tie region. *From B4.*

## 3. Track B, one line each

- **B1 — imbalanced baselines** (312 runs). Focal is the only competitive member. The pattern is *dataset*-driven: on DermMNIST focal matches or slightly beats TraLO's macro-F1; elsewhere TraLO leads or ties. None of the three satisfies the cap without the post-hoc LP.
- **B2 — native resolution** (96 runs, native HAM10000). The macro-F1 margin over clipping reproduces: **+0.013, positive in all 6 backbone×cap cells**. cc-F1 ties both duals. Satisfaction: TraLO 24/24, Hounie 24/24, Fioretto 0.96, LP-LG 0/24.
- **B3 — ALM** (24 runs). *The strongest new result.* ALM is genuinely a stronger dual than Fioretto (beats it by +0.010), and TraLO still tops it in **all six** tight-cap cells by +0.028 cc-F1.
- **B4 — tie-region ablation** (20 runs). The reset and hinge carry the *tight-cap* advantage specifically, not a general one. Consistent with the mechanism: they bind only where the cap binds.

- **B5 — win-bar sensitivity** (no new runs). The dual win at L30/L40 is stable at every threshold (4/0/0 even at the strict $\tau$=0.010). The clip comparison never produces a stable win at any threshold.
- **B6 — bootstrap CIs** (no new runs). 20 000 resamples on the tight-cap cells.
- **B7 — BH-FDR** (no new runs). 3 of 8 cells survive against Fioretto; 0 of 8 against the clipper.
- **B8 — MobileNetV2** (32 runs). The advantage reproduces on a fourth backbone.

**The through-line, if he asks what Track B established**

The advantage is *specific*, and the follow-ups sharpen where. Against the **dual-ascent family** at tight caps it is real, survives a stronger dual (ALM), reproduces on a fourth backbone, and holds at every win threshold. Against **post-hoc clipping** the durable, categorical separation is *native cap satisfaction* (1.0 vs 0.0 everywhere) plus a macro-F1 margin that now also reproduces at native resolution. The honest boundary: on the imbalance-friendly dataset a focal clipper closes the macro-F1 gap. That boundary is now stated in the paper rather than left open.

## 4. The three imbalanced-learning baselines (B1)

First, the framing point, because it is the one most easily got wrong: **these are not clipping algorithms.** They are *training losses* — three ways of making a classifier care more about a rare class while it learns. Clipping is a separate step applied afterwards. That is exactly why they belong in the paper: none of the three controls *how many* items get the constrained label, so each still needs the post-hoc LP to meet the cap. Their presence tests whether a smarter *training* recipe plus clipping can match constraint-time training.

All three run at their original papers' settings, verified against the source texts.

**Focal loss** `lin2017focal` — **Lin et al., ICCV 2017**

$$\mathcal{L}_{\text{focal}} \;=\; -\,\alpha\,(1-p_y)^{\gamma}\,\log p_y, \qquad \alpha = 0.25,\;\; \gamma = 2$$

where $p_y$ is the predicted probability of the *true* class.

- $\gamma$ is the **focusing exponent**. It decides how fast already-correct examples stop mattering. At $p_y$=0.9, the factor $(1-0.9)^2 = 0.01$ — that example contributes 100× less than under plain CE. At $p_y$=0.5 it is only 4× less.
- $\alpha$ is a fixed scale on the whole term.

**Why it suits imbalanced data:** the majority class is learned early and becomes "easy". Under plain CE those easy examples still dominate the gradient by sheer count. Focal drains their contribution so learning stays pointed at what is still hard — which is disproportionately the rare class. Note it reweights by *difficulty*, which correlates with rarity but is not the same thing.

**Class-balanced reweighting** `cui2019class` — **Cui et al., CVPR 2019**

$$E_{n_c} \;=\; \frac{1-\beta^{\,n_c}}{1-\beta} \qquad\Longrightarrow\qquad w_c \;\propto\; \frac{1}{E_{n_c}} \;=\; \frac{1-\beta}{1-\beta^{\,n_c}}, \qquad \beta = 0.9999$$

with $n_c$ the number of *training* samples of class $c$; weights are normalised to mean 1 so the loss scale still matches CE.

- $\beta$ sets a **saturation scale** of $1/(1-\beta)$ = 10,000 samples. Below that, a class is weighted close to inverse-frequency; far above it, extra samples stop adding weight.

**Why it suits imbalanced data:** the naive fix, weighting by $1/n_c$, over-corrects. Cui et al.'s argument is that samples overlap — the 40,000th image of a class teaches almost nothing new — so the useful quantity is an *effective* number, not a raw count. **Why it is active on our data:** OctMNIST's training classes run from roughly 7.7k to 46k, straddling the 10k saturation scale, so some classes sit below the saturation point and others well above — the weights genuinely differ.

**Logit adjustment** `menon2020long` — **Menon et al., ICLR 2021**

$$\mathcal{L}_{\mathrm{LA}} \;=\; \mathrm{CE}(z + \tau \log \pi,\; y)\,, \qquad \tau = 1$$

where $z$ are the logits and $\pi$ the training class prior.

- $\tau$ scales the prior correction. $\tau{=}1$ is not a tuned knob: it is the value that makes the classifier Bayes-optimal for *balanced* error. Anything else would need justifying.

**Why it suits imbalanced data:** a model trained on a skewed set learns the training prior along with the signal. Adding $\tau \log \pi$ to the logits *during* training handicaps the frequent classes, so the network must earn its confidence on them instead of coasting on base rates. It shifts the decision boundary rather than reweighting anything.

**Why these three, and not three flavours of one idea**

They attack three genuinely different mechanisms: focal reweights per *example* by difficulty; class-balanced reweights per *class* by frequency; logit adjustment moves the *decision boundary* by the prior. That makes them a fair panel rather than a stacked one. **If asked whether they were handicapped:** they run at their authors' own settings, and focal *beats* TraLO on DermMNIST — these are not baselines set up to fail.

## 5. Reading the statistics (B5–B7)

Three different ways of interrogating the same question: *is the OctMNIST tight-cap cc-F1 result real, or an artifact of a small sample?* Each guards against a different failure. All three are re-analyses — no new runs.

### The underlying problem: $n = 4$ seeds

Four seeds per cell is very few, and it creates two separate worries. *Is this one number reliable?* (B6.) *I ran many comparisons — how many won by luck?* (B7.) Fixing one does not fix the other. B5 adds a third: *does my conclusion depend on where I drew the line?*

### B5 — win-bar sensitivity

The paper calls a cell a "win" when the gap clears +0.005. That threshold is a judgement call, so B5 re-scores everything at $\tau \in \{0.003, 0.005, 0.010\}$. **Result:** against Fioretto the tight-cap win is 4/0/0 at *every* threshold, including the strict 0.010. Against the clipper it is never stable at any threshold. **What it protects against:** threshold-shopping. A conclusion that flips when you nudge the bar is a property of the bar, not of the data.

### B6 — bootstrap confidence intervals

With $n{=}4$ you cannot lean on a $t$-test's normality assumption. Instead, resample the paired differences *with replacement* 20,000 times, recompute the mean each time, and take the middle 95% of those means. That range is the interval — an answer to "how much would this number move on a slightly different sample?" without assuming a distribution. The resampling unit is *cells*, not seeds, which respects the atomic-unit rule.

| comparison | cap | mean | 95% CI | reading |
|---|---|---|---|---|
| TraLO − Fioretto | L30 | +0.046 | [+0.024, +0.074] | entirely above 0 |
| TraLO − Fioretto | L40 | +0.033 | [+0.021, +0.044] | entirely above 0 |
| TraLO − ALM | L30 | +0.035 | [+0.010, +0.048] | entirely above 0 |
| TraLO − clip | L30 | −0.003 | [−0.013, +0.008] | **straddles 0** |
| TraLO − clip | L40 | +0.002 | [−0.010, +0.012] | **straddles 0** |

The specific cell reviewers flagged as weak — ViT-B/16 at L30, about $1.8\sigma$ — comes out at +0.086 against Fioretto and survives. Note also the *shape*: at L10/L15/L20 even the dual gap ties, because the tightest caps ceiling-crush every method. The win is a mid-cap phenomenon.

### B7 — Benjamini–Hochberg FDR

Run 20 tests at $p < 0.05$ with nothing real going on and you should *expect* about one "significant" result. BH controls the **false discovery rate**: among the results you declare significant, at most $q$=5% are expected to be false. Mechanically, the bar gets stricter the more tests you ran. **Result:** 3 of 8 cells survive against Fioretto; **0 of 8** against the clipper.

### Why "3 of 8" is not the disappointment it looks like

With 4 seeds, a within-cell paired Wilcoxon test **cannot return** $p < 0.125$ — there are not enough orderings. That is a floor imposed by the sample size, not by the effect. Most cells therefore cannot reach individual significance however large the true effect is. So the meaningful quantity is not 3/8 in isolation; it is **3 versus 0**.

**What "0 of 8 on clipping" does and does not mean — know this exactly**

It means: *on cc-F1, at OctMNIST tight caps, TraLO does not beat the post-hoc clipper.* The paper reports that comparison as a **tie**, and B5, B6 and B7 are three independent reasons why.

It does **not** mean TraLO fails to beat clipping in general. The deployment claim — overall quality, macro-F1, +1.6 to +5.3 pp, positive in 24 of 27 cells — is a *different metric* from a *different analysis*, and it stands. So does native cap satisfaction, 1.0 vs 0.0, which is categorical.

Do not let these two get conflated in the room. B5–B7 are about *cc-F1 at tight OctMNIST caps* only.

### Which of B5–B7 actually reached the manuscript

Worth knowing precisely, because two are in and one is not.

| | in the paper? | where |
|---|---|---|
| **B6** bootstrap CIs | **yes**, in blue | §5.1 check *(vi)*, `main.tex:746` — the +0.046 / +0.033 gaps with their CIs |
| **B7** BH-FDR | **yes**, in blue | same sentence, `main.tex:749` — "three of the eight ... and none against the post-hoc clipper" |
| **B5** win-bar sensitivity | **no** | — |

B6 and B7 sit in a single long sentence, which is why they can read as one result rather than two. The other Benjamini–Hochberg mentions (`main.tex:637-645`) are *pre-existing black text*, not B7.

**If he asks why B5 was left out**

Not because it was negative — it is one of the more supportive results in the campaign: the tight-cap win is 4/0/0 against Fioretto-LDF at *every* threshold tested ($\tau \in \{0.003, 0.005, 0.010\}$), and the clip comparison is never stable at any of them. The likely reason is **redundancy**: the paper already argues threshold-robustness at `main.tex:802`, by a different route — that the cell-mean gaps range +0.016 to +0.081, so no plausible threshold changes the classification. That sentence is the professor's original text.

**The case for adding it:** B5 upgrades that argument from "the gaps are large, so the threshold cannot matter" to "we re-scored at three thresholds and the classification did not change" — a directly tested claim rather than an inferred one. One clause, in blue, and it strengthens an existing sentence rather than introducing a new result.

**Where the paper already says this — exact lines to point at**

The paper uses **two metrics with two different comparators**, deliberately: **cc-F1 is measured against the trained duals; macro-F1 against the clipper.** If he asks where you concede the clip tie:

- `main.tex:651` — the section is titled "Constrained-class quality *vs. the trained baselines*".
- `main.tex:659` — "the package leads *the best constraint-trained baseline* on every backbone, by a mean cc-F1 gap of +0.038".
- `main.tex:708` — stronger than a tie, an explicit concession: "Where a *post-hoc clipper posts higher raw cc-F1* (three of the six band cells, by up to +0.018), the edge is bought by editing ... TraLO leads that same cell 0.702 to 0.666 on macro-F1".
- `main.tex:677` — Figure 1 footnote: "faded post-hoc clippers can post higher raw cc-F1 by editing".
- `main.tex:1764` — appendix column headers: "Δcc-F1 vs. *trained*" and "Δmacro-F1 vs. *clipper*".

So on cc-F1 at tight caps the clipper wins 3 of 6 cells outright. The paper's answer is not to deny it but to show *how* it is bought — clipping rewrites roughly 11% of the batch — and that TraLO wins the same cell on macro-F1.

**One wording fix worth raising yourself**

The abstract states this result *without naming the comparator*: "one regime shows a consistent advantage across all backbones tested: OctMNIST at tight caps". Read cold that scans as an advantage over *everything*; the scoping only arrives in §5.1. Adding three words — "advantage *over the trained baselines*" — removes the ambiguity. Better to propose this than to have a reviewer read it the loose way.

**Where this helps you.** A null result you found yourself and reported is a much stronger position than one a reviewer finds for you. You tested the clip comparison three ways, it did not hold, and the paper says so — which is also why the headline is anchored on the dual-ascent comparison, where all three methods agree the effect is real. It is consistent with the July quota-fill finding: the apparent cc-F1 edge over clipping was a budget-fill artifact, and the durable separation from clipping is native satisfaction, not a quality margin on that metric.

**Warning: two different B6/B7 analyses exist in the repo**

`results/stats_trackb/b6/` and `b7/` are an *older, broader* sweep covering EfficientNetB0, ResNet18, aider and eurosat — backbones and datasets not in the current grid — reporting 259/270 wins surviving FDR. Those are **not** the paper's numbers. The B6/B7 above are the Track-B ones, scoped to the OctMNIST tight-cap family.

## 6. Numbers to know cold

| Quantity | Value |
|---|---|
| Main grid | 1944 runs, 3 datasets × 3 backbones × 9 caps × 4 seeds |
| Track B added | ≈600 runs across B1–B8 |
| Headline macro-F1 gain over clipping | +1.6 to +5.3 pp (dataset means) |
| OctMNIST tight-cap cc-F1 lead vs duals | +0.038 over 6 cells, sign test $p$=0.031 |
| TraLO − ALM, tight caps | +0.028 cc-F1, 6/6 cells |
| ALM − Fioretto | +0.010 cc-F1 (so ALM is the tougher baseline) |
| Native HAM10000, macro-F1 vs LP-LG | +0.013, 6/6 cells positive |
| Native satisfaction (TraLO / LP-LG) | 24/24 vs 0/24 |
| Ablation: reset alone / hinge alone / both | +0.079 / +0.032 / +0.036 (not additive) |
| Atomic unit of analysis | (dataset, backbone, cap) over 4 seeds — never pooled |

## 7. What changed last night, and why

An audit of the full 32 pages extracted 875 checkable assertions and found five real problems. Two were acted on.

### Four stale disclaimers

Four passages written *before* Track B existed now contradicted the blue work — the paper said it had not done things it now does. Each is struck in red with a blue replacement:

- abstract, intro, §7: "native-resolution generalization is left to future work"
- §2: "we do not run a dedicated augmented-Lagrangian baseline" (ALM is now Table 2)
- §2 and §7: the imbalanced comparison described as not run (it is now §5.3)

These are all cases where the new work *strengthens* the paper and the old text had not caught up.

### Table 3 lost a column

Table 3 had a `vs. LP-LG` column taken from the B1 campaign's own clip arm. Table 1 reports the same quantity from the main corpus, and the two disagreed in all nine cells — including a sign flip, and +0.001 where Table 1 says +0.081.

> **If he asks about this, the answer is short**
>
> Two different run sets were being asked the same question. The column was deleted, not recomputed — substituting corpus numbers into a B1 table would mix run sets inside one row, which is worse. TraLO-versus-clipping belongs in Table 1 and Appendix D, over the full grid. Note the direction: *the deleted column understated the result.* Table 1's numbers are the larger ones.

### Table 4's bolding was fighting its own caption

The rule was "best in row", so Fioretto took bold in 4 of 6 macro-F1 rows — on margins of 0.001–0.002, i.e. *inside* the ±0.005 band the paper calls a tie. The table crowned a winner where the caption said there was none. Bold now marks everything within 0.005 of the row best, which leaves **LP-LG as the only unbolded method in all six macro-F1 rows** — the claim made visible. The caption also said "every cell is a tie" against the duals; one cell (MobileNetV3, $L$40) leads Hounie-RCL by +0.008, a TraLO win that was being under-reported.

### The ablation table (Table 5) — two corrections

An independent reviewer re-derived four findings that were already sitting unapplied in `docs/archive/AUDIT_` Two were table-level defects and are now fixed in red/blue:

- **The "Undershoot hinge" row never isolated the hinge.** `tralo_bounded/train.py` contains *zero* occurrences of the optimizer reset, so TraLO − TraLO-bounded removes *both* components. The table's own numbers give it away: reset alone costs +0.079, reset+hinge costs +0.036 — removing more cannot hurt less. Row relabelled **"Reset + hinge (TraLO-bounded)"**, and a new row **"Undershoot hinge (alone)"** was added underneath it, from 24 runs dispatched and completed on 2026-08-13 — so the hinge is now isolated on the very cells that carry the claim. It costs +0.032 cc-F1, winrate 0.92, positive in all six cells.
- **The caption called itself "Complete" and was not.** `ablation_complete.csv` has five arms at $n$=24; the table showed four. The omitted one is `+KL` (mean −0.0103, $p$=0.0398) — significant, but it is an *addition*, not a removal, and it adds a term the paper does not use ($\alpha_{\mathrm{KL}}$=0 corpus-wide). Rather than show a row for a variant that is not the method, the word "Complete" was struck from the caption, and the orphaned KL sweep paragraph in Appendix B — which described the same dead term — was struck with it. *If he asks why the KL result is not shown: because it argues about a hyperparameter we set to zero, and the honest fix is to stop claiming completeness, not to advertise an arm we do not run.*

Also fixed: Table 9's threshold language (it called +0.002 "a positive margin" while calling −0.003 elsewhere "no systematic advantage"), and the "eight tight-cap cells" phrasing, which now says *which* eight.

**Bibliography**

51 citations, 55 entries, zero undefined. Fixed: the Cooper library reference was dated 2024 but was posted April 2025; missing diacritics in Pathak et al.; and ALM had been introduced with no citation at all (now Bertsekas 1982). The two *added* entries also render blue in the reference list — the `.bib` carried no colour, so they were appearing black even though their in-text citations were blue. The two metadata *corrections* above are not marked, since they modify pre-existing entries.

## 8. Questions he is likely to ask

**"Did you touch my text?"**

No, and it was checked mechanically rather than by eye. Struck sentences are still character-for-character his — only a wrapper was added around them, which the checker confirms.

**"Why is the figure caption entirely blue?"**

The dataset figure moved from §1 to Appendix E. An image cannot be colour-marked, so the relocation is signalled by making its caption blue. His wording is preserved almost verbatim; one clause was added. *This is a judgement call — you may prefer to blue only the new clause and leave his caption text black.*

**"Does focal beating you on Derm undercut the paper?"**

No, and it is stated in the paper now rather than buried. It narrows the claim from "constraint-trained beats clipping" to "beats *vanilla* clipping" on that one dataset. The count-satisfaction separation is untouched: none of the three imbalanced recipes meets the cap without the post-hoc LP.

**"Is the cc-F1 win real?"**

Against the duals, yes — and it is the claim with the most support behind it: stable at every win threshold (B5), surviving a stronger dual (B3), reproducing on a fourth backbone (B8), FDR-significant (B7). Against the clipper it is *not* claimed as a win, and B5/B7 say so explicitly (0 of 8 cells survive FDR). The paper reports the clip comparison as a tie.

**"Is the hinge doing anything on its own, or only alongside the reset?"**

On its own. This was the one genuine hole in the paper's own framing, and it was closed the night before this meeting with 24 fresh runs.

The arm is `hybrid_mode=bounded_only` with `reset_optimizer_at_sat=True` — the hinge removed, the reset *retained* — on exactly the grid Table 5 uses: OctMNIST $\times$ $\{L30, L40\}$ $\times$ {MobileNetV3, RegNetY400MF, ViT-B/16} $\times$ 4 seeds. All 24 completed, zero failures.

| Backbone | $L30$ | $L40$ |
|---|---|---|
| MobileNetV3 | +0.013 | +0.016 |
| RegNetY400MF | +0.048 | +0.009 |
| ViT-B/16 | +0.052 | +0.056 |
| *overall* +0.032, helps in 22 of 24 runs, 6/6 cells positive | | |

Every cell is positive, so the hinge clears the same cell-level sign test as the reset ($p$=0.031, the floor at six cells). **Both** named components are now load-bearing individually, which is what the title claims.

The one number worth being ready for: reset alone +0.079, hinge alone +0.032, both together only +0.036. That looks wrong until you say it out loud — *the two are not additive*. They fix overlapping failure modes, so removing both costs less than the sum of removing each. The caption now says this explicitly rather than leaving him to notice it.

## 9. Open items — know these before he raises them

- **The "1.6–5.3 pp" range pools across backbones and caps.** Each endpoint is a dataset mean over $3 \times 9 \times 4$ runs. Appendix D's own stated rule is that the atomic cell is (dataset, backbone, cap) over seeds *only*. At that granularity the same quantity ranges from −0.018 to +0.091. The headline is defensible as a summary but it is not scoped the way the appendix says to scope things. *Unresolved — your call.*
- **Table 1's caption says "the full symmetric grid"** but shows 3 of 9 caps (27 of 81 cells), and the L40 number quoted beside it is not in the table.
- [CLOSED 13 Aug] **Only one of the "two portable components" was isolated on the cells that carry the claim.** This was the likeliest place for a reviewer to press. The missing 24 runs were dispatched and completed, and they settled it *in the paper's favour* — numbers in §8.5 above.
- **OctMNIST has no native-resolution counterpart**, so whether the tight-cap cc-F1 win survives at full resolution is untested. Stated plainly in the paper.
- Roughly 70 further minor findings from the audit were never adversarially verified — the run hit its limit. Treat them as unchecked leads, not defects.

**The one you should expect him to raise: the OctMNIST headline**

Two independent reviews — the July audit and a fresh one — have questioned whether the +0.038 cc-F1 lead at OctMNIST tight caps is a *quality* advantage or a *budget-utilisation* one. The argument runs: precision is flat across the three trained methods, so the cc-F1 ordering tracks how much of the cap each method spends. The paper already describes this mechanism in its own words (§5.1: the duals "overshoot it downward and finish 11–36% under the budget at $L30$ ... so recall falls with the unspent budget while precision holds") and frames it as TraLO recovering the leftover headroom.

A reframing was drafted and **withdrawn at the author's direction**; the abstract and §5.1 read exactly as the professor wrote them. Nothing in the paper currently states the budget-utilisation reading.

**If he raises it**, the facts to have ready: the decisive control — refill each dual's unspent budget with its next-most-confident predictions and re-score — has not been run. And the claim is scoped to the trained duals throughout (§5.1's title and first sentence), not to clipping, which B5–B7 already report as a tie. The supporting dossier is `docs/archive/AUDIT_2026-07-31.md`.

## 10. Files, if he wants them

| | |
|---|---|
| `paper/main.tex` | his original, untouched |
| `docs/main.tex` | the blue revision (source) |
| `paper/main_rev.pdf` | the blue revision, 32pp — **the one to read** |
| `paper/references.bib` | 55 entries, 51 cited |
| `docs/track_b/TRACK_B_RESULTS_HANDOFF.md` | full B1–B8 record |

Committed as `8648b15e` on branch `headroom/small-cnn`. Still to push to `origin/tmlr/submission`.